

Linear Optimization

Modeling and Solving Linear problems with Julia
and WebCAS

Peter Staab
Fitchburg State University

September 9, 2025

Contents

Preface	iv
1 Background Mathematics for Linear Optimization	1
1.1 Lines and Planes	1
1.2 Using Software for Plotting	11
1.3 Matrices and Matrix Reduction.	15
1.4 Using Software to Perform Matrix Pivots	30
2 Developing Linear Models	36
2.1 Introduction to Linear Optimization	36
2.2 Linear Optimization Problems	44
2.3 Geometry of Feasible Sets.	47
3 The Simplex Method	53
3.1 Slack Variables and Dictionaries	53
3.2 Tableaus and the Simplex Method.	60
3.3 Simplex Method for Infeasible Tableaus.	68
3.4 Infeasibility and Unboundedness	74
3.5 Summary of the Simplex Method	77
4 Integer Linear Optimization Problems	81
4.1 Introduction to Integer Linear Optimization Problems	81
4.2 Cutting Planes.	84
4.3 Branch and Bound	96
5 Solving Linear Optimization Problems using Software	108
5.1 Introduction to the Julia Language	108
5.2 Vectors and Matrices in Julia	111
5.3 Linear Algebra in Julia.	116
5.4 Julia Packages for LOPs	118
5.5 Cutting Planes and Branch and Bound with Julia	124
6 Dual Problems	130
6.1 Introduction to Dual Problems	130

6.2	Primal-Dual Relationships	135
6.3	Finding Optimal Certificates.	138
7	General Linear Optimization Problems	141
7.1	General Form of an LOP	141
7.2	Solving General Linear Optimization Problems	148
7.3	General Dual Problems.	154
7.4	General Applied Problems	154
8	Matrix Formulation of Linear Optimization Problems	163
8.1	Writing a LOP in Matrix Form	163
8.2	Performing LOP Matrix Operations using Software	168
8.3	Performing the Matrix Simplex Method.	174

Appendices

A		178
B	Installing Julia	179
C	Managing Packages in Julia	180

Back Matter

Chapter 1

Background Mathematics for Linear Optimization

1.1 Lines and Planes

Objectives

- Write a line in **intercept form** and use the line to produce a plot.
- Plot the set of points defined by a set of inequalities and define this as a **feasible set**.
- Determine if a feasible set is bounded or unbounded.
- Define a plane and hyperplane as a linear equation.
- Define a convex set.

I'm guessing that if you are reading this book (either you picked it up or have been assigned it) that you have a fair bit of mathematical background. That being said, I'm guessing that you are quite familiar with lines—you may even be embarrassed if you are seen reading a section on lines. However, review is nearly always good and seeing the context of old material in a new way expands your thinking. In short, it will be worth your while to read through this chapter.

1.1.1 Lines

When most mathematics students are asked about lines, the response is $y = mx + b$, the **slope-intercept form** of the line. This is often presented in a Precalculus class in which linear functions fall from this naturally. However, not all lines can be written in this form. The exception is vertical lines, like $x = 1$.

Instead, we will use the **general form** of the line or

$$ax + by = c \tag{1.1.1}$$

as long as a and b are both not zero,¹ and the b in this formula is not the y -intercept. This form allows horizontal, vertical and any oblique lines.

¹A nice way to ensure this is $ab \neq 0$.

This form is quite nice in that if a is zero, then the line is horizontal, if b is zero, then the line is vertical and if both are non-zero then the line is oblique with x -intercept of c/a , y -intercept of c/b and slope of $-a/b$. If we take the line in (1.1.1) with $ab \neq 0$ and divide through by c , this can take on the **intercept form of the line**:

$$\frac{x}{x_0} + \frac{y}{y_0} = 1 \quad (1.1.2)$$

and in this form the x -intercept is x_0 and the y -intercept is y_0 .

Example 1.1.1 Put the line given by $3x + 5y = 30$ into intercept form and find the intercepts.

Solution. Divide through by 30 to get

$$\frac{3x}{30} + \frac{5y}{30} = 1 \quad \text{or} \quad \frac{x}{10} + \frac{y}{6} = 1$$

This shows that the x -intercept is 10 and the y -intercept is 6. □

The above example showed how we can use the intercept form to easily read off the intercepts. The following exercise will use these to plot the lines.

Checkpoint 1.1.2 Plot the following lines: $x = 3$, $y = 4$, $2x + 3y = 24$ and $-3x + 4y = 12$ on the same axes. Note: do this by hand instead of graphing calculator/website/software.

Hint. Use the information above to note if it is horizontal, vertical or oblique and in the latter case, use intercepts to plot the line.

Solution. In this case, note that the first line is vertical, the second is horizontal. For the third line, divide through by 24 and simplify to

$$\frac{x}{12} + \frac{y}{8} = 1$$

which will have the intercepts (12,0) and (0,8) and the last line can be simplified to

$$-\frac{x}{4} + \frac{y}{3} = 1$$

which will have the intercepts (-4,0) and (0,3).

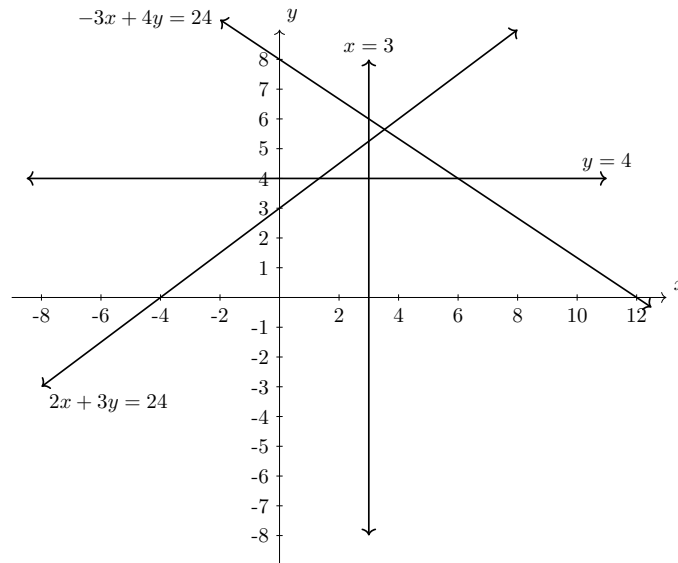


Figure 1.1.3 A graph of the lines $x = 3$, $y = 4$, $2x + 3y = 24$ and $-3x + 4y = 12$ with the result shaded.

1.1.2 Linear Inequalities

Related to lines is that of linear inequalities. Consider the inequality $3x + 4y \leq 12$ and let S be the set of all points that satisfy the inequality. The line $3x + 4y = 12$ cuts the xy -plane into two regions and in this case, the line is included in the set S .

Which side of the line satisfies the inequality? Since every point that is not on the line either satisfies the inequality or not, we can pick any point to determine this. For example in the line $3x + 4y = 12$ above, an easy point to pick is $(0,0)$ and since $3(0) + 4(0) \leq 12$ is true, then the side of the line containing the origin is in the set. The following is plot:

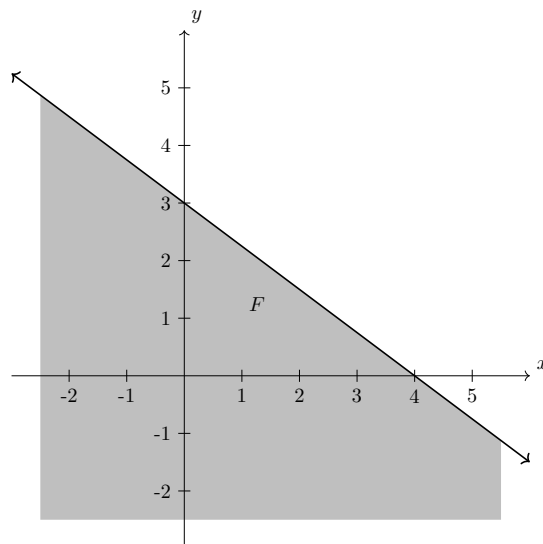


Figure 1.1.4 A plot of the inequality $3x + 4y \leq 12$ and the shaded part is the set of points that satisfy the inequality.

The set that satisfies the inequality we have labelled F and shaded in light

gray. When combined with other inequalities, we often call the set a **feasible set**, hence labelling it as F .

As we will see throughout this book, it is common that there are multiple inequalities. For example, let's add the inequality $y \geq x + 1$ to the inequality above and determine the set of points that satisfy both inequalities. There are two techniques that can be used.

Technique 1: Graph the lines and Pick. First, let's plot the two lines (ignoring the inequality) $y = x + 1$ and $3x + 4y = 12$. The first line is in slope-intercept form and the second was plotted above. A plot with both graphs is

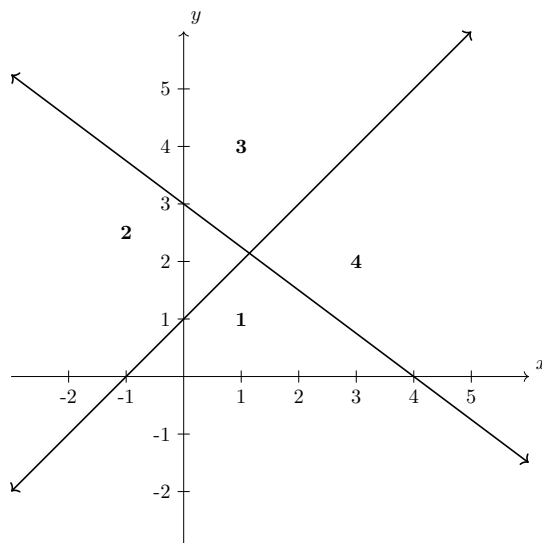


Figure 1.1.5 A graph of the lines $y = x + 1$ and $3x + 4y = 12$, which cuts the plane into four regions. These are numbered 1 through 4.

The 2 lines split the plane into four regions and only one of these satisfies both inequalities. If we pick four points (one for each region), we can determine the feasible set. The following table does this formally.

Table 1.1.6 Evaluation of inequalities

Region #	Point	$3x + 4y \leq 12$	$y \geq x + 1$
1	(0,0)	true	false
2	(0,2)	true	true
3	(1,4)	true	false
4	(3,2)	false	false

From [Table 1.1.6](#), region 2 satisfies both inequalities. Therefore this is the feasible set together with the lines that are the boundary of region 2.

This technique will work however, as the number of lines increases it become increasingly difficult to plot and determine all of the inequalities. The next technique is often more desirable.

Technique #2: Cross Out Regions Instead, we start with one region and then cross out the region not in the feasible set. Each subsequent inequality, we repeat and the only region not crossed out is the feasible set. Performing this technique with the same pair of equalities, we'll start with the inequality $3x + 4y \leq 12$. We've already seen this and will cross out the region in the northeast part.

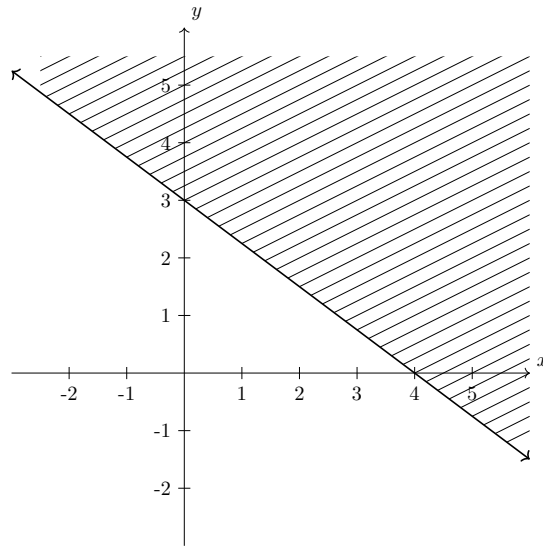


Figure 1.1.7 A graph of the line $3x + 4y = 12$ and the side of the line that does not satisfy the inequality $3x + 4y \leq 12$ is crossed out.

Now we will add the second inequality, $y \geq x + 1$. Graph the line and then cross out the region that does not satisfy the inequality. Note that this is the southeast side of the line.

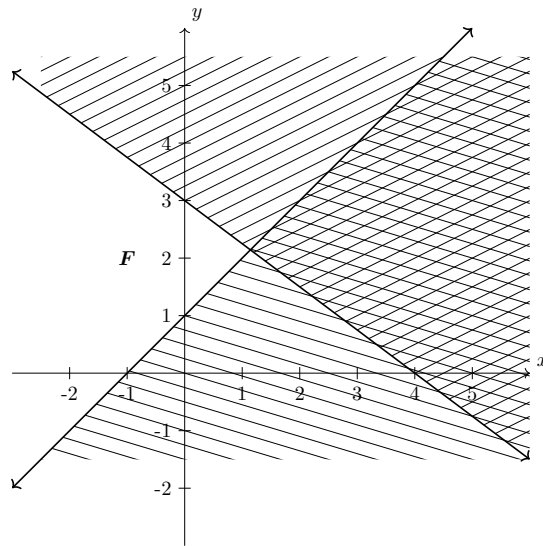


Figure 1.1.8 A graph of the lines $3x + 4y = 12$ and $y = x + 1$ with the corresponding inequalities crossed out. The feasible set is labelled **F**

From [Figure 1.1.8](#), the region not crossed out is the feasible set and is the same that was found with Technique #1 as shown in [Figure 1.1.5](#). Although either technique will result in the correct answer, this one is generally faster to perform with a large number of inequalities.

Checkpoint 1.1.9 Find the feasible set for the following set of inequalities

$$\begin{aligned} x &\geq 0, \\ y &\geq 0, \\ x + y &\geq 6, \end{aligned}$$

$$5x + 8y \geq 40.$$

Solution. In this case, we'll graph and cross out the false side of each line. The first two lines are the coordinate axes. The 3rd and 4th lines can be written in intercept form as

$$\frac{x}{6} + \frac{y}{6} = 1 \quad \frac{x}{8} + \frac{y}{5} = 1$$

So the 3rd line has the intercepts $(6, 0)$ and $(0, 6)$. The 4th line has the intercepts $(8, 0)$ and $(0, 5)$.

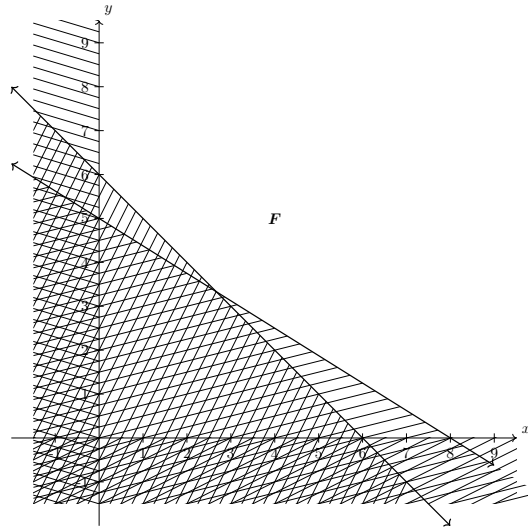


Figure 1.1.10 The feasible set of the given inequalities. The side of the line not in the feasible set is crossed out and the remaining region is the feasible set and labelled F .

and from the plot, the region in the northeast section that is not crossed out is the feasible set.

Note: as we will see throughout this book, it is common that we have a nonnegative constraint, like $x \geq 0$ and $y \geq 0$. With only two variables, this means outside the first quadrant is crossed out.

1.1.3 Bounded and Unbounded Feasible Sets

Another feature of feasible sets is that of being bounded. For feasible sets in the plane, a feasible set that is a polygon (or technically the interior of a polygon), is bounded. For example, in [Section 2.3](#), we will see the feasible set F be set defined by

$$\begin{aligned} x + y &\leq 7, \\ 2x + 3y &\leq 18, \\ x &\geq 0, \\ y &\geq 0. \end{aligned}$$

which can be plotted as

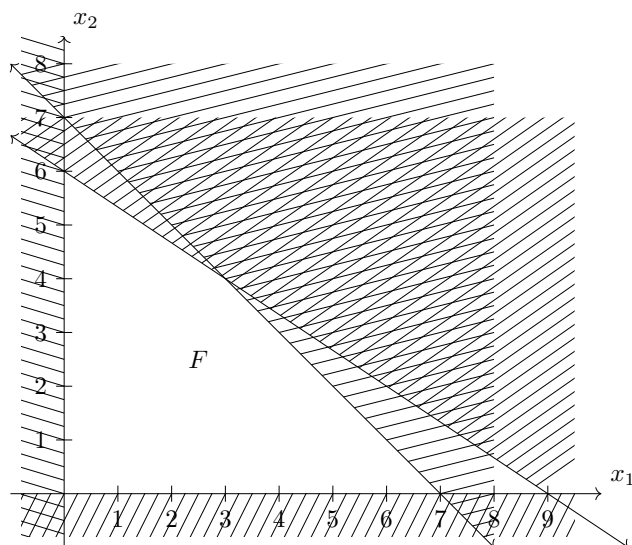


Figure 1.1.11 A graph of the feasible set of the above problem. For each inequality, the side of the line not in the feasible set is crossed out. The set of points left is the feasible set for the inequalities and labelled F .

The feasible set is the interior of the polygon together with its boundary and appears to be bounded. In contrast, the feasible set in [Checkpoint 1.1.9](#) is not a polygon and the feasible set extends without bound. The definition below is more general for a feasible set in any dimension.

Before defining a bounded feasible set, we need the following:

Definition 1.1.12 A ball in $\mathbb{R}^n$ centered at the origin is the set points

$$B = \{(x_1, x_2, \dots, x_n) \mid x_1^2 + x_2^2 + \dots + x_n^2 \leq r^2\}$$

◇

A ball in $\mathbb{R}^2$ is the circle with its interior. In $\mathbb{R}^3$, it is a sphere with its interior.

With the knowledge of a ball, we can now define a bounded feasible set.

Definition 1.1.13 A feasible set, F in $\mathbb{R}^n$ is **bounded**, if there exists a ball centered at the origin, B such that $F \subset B$. If no ball exists, the set F is **unbounded**. ◇

The feasible set in [Figure 1.1.11](#) is bounded because the ball with radius 8 would encompass the feasible set. The feasible set in [Checkpoint 1.1.9](#) has no ball that encompasses it, so is unbounded.

1.1.4 Planes and Hyperplanes

A linear equation in three variables is a **plane**. For example the equation $x + 2y - 3z = 12$ describes points in $\mathbb{R}^3$, the set of all three-dimensional space. Like a line in $\mathbb{R}^2$, it extends indefinitely and is flat. The following shows a plot of the plane $2x + 3y + 4z = 24$.

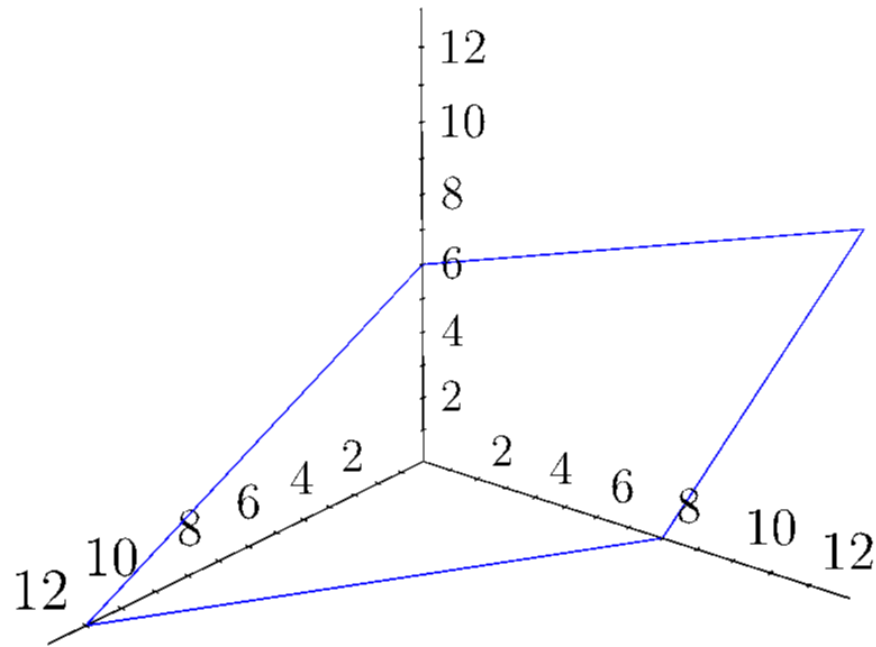


Figure 1.1.14 A plot of the plane $2x + 3y + 4z = 24$ in $\mathbb{R}^3$. If reading on the web, this figure is interactive and the orientation can be changed.

We will encounter inequalities involving three variables. As seen in two dimensions, the feasible set will be the set of all points satisfying all of inequalities. It is difficult to graph this, but one can imagine that in three dimensions, the same ideas hold as those above. For example, one can cross out the side of the plane that is not in the feasible set and then the region in $\mathbb{R}^3$ is the region not crossed out.

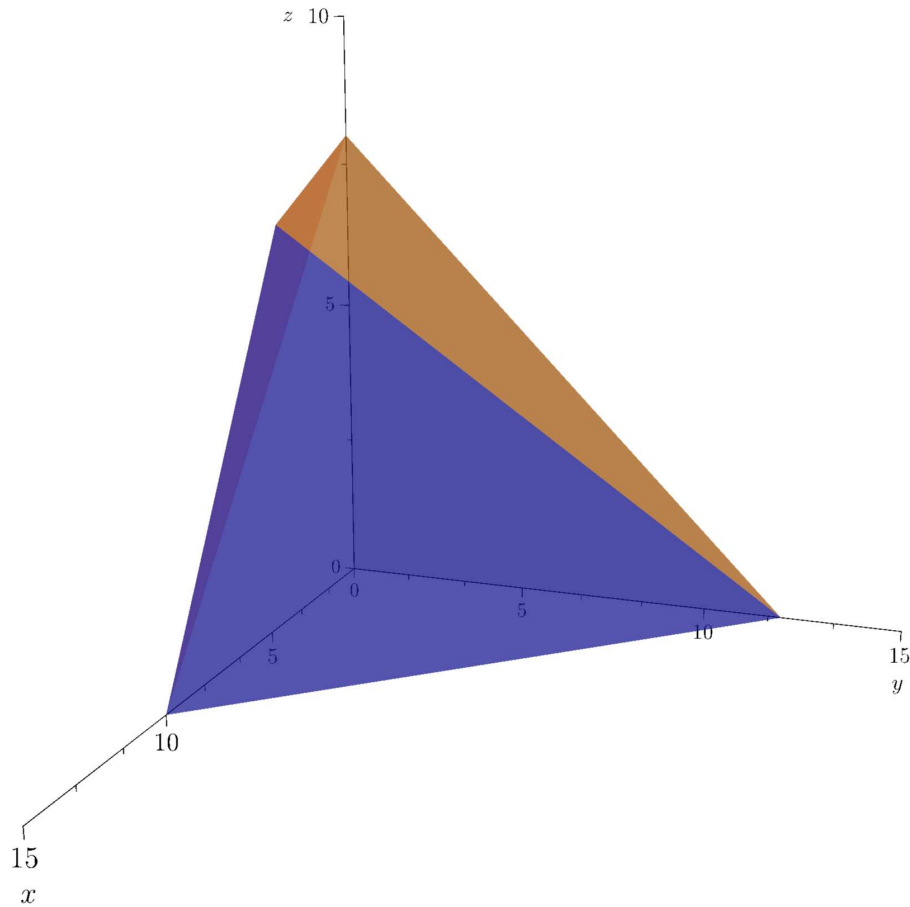


Figure 1.1.15 A plot of a feasible set of in $\mathbb{R}^3$. If this is being viewed in a web browser, the graph should be interactive in that it can be rotated.

Linear equations with more than 3 variables are often called **hyperplanes**. Although impossible to graph, these work the same ways as lines and planes in that they cut the Euclidean space in half.

1.1.5 Convex Sets

As we will see later, a convex feasible set is key to the simplex method working. What does *convex* mean? We will explore this and show that a feasible set constructed of linear inequalities is always convex.

Definition 1.1.16 A set S in $\mathbb{R}^n$ is **convex** if for any two points $\mathbf{x}$ and $\mathbf{y}$ in S that all points in the line segment connecting $\mathbf{x}$ and $\mathbf{y}$ is in S $\diamond$

Note 1.1.17 Note: a line segment between points $\mathbf{x}$ and $\mathbf{y}$ can be written

$$L = \{\mathbf{x}t + \mathbf{y}(1-t) \mid t \in [0,1]\}.$$

In two dimensions, a convex set is a set in $\mathbb{R}^2$ such that its boundary is convex in the standard sense. Sets like the interior of circles, rectangles and the “inside” of parabolas are examples. To get a sense of this, the sets consisting of the interior of the following objects are convex:

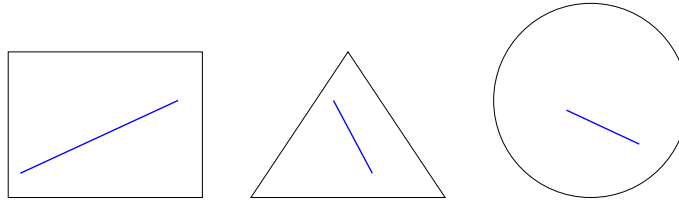


Figure 1.1.18 Examples of convex sets (interiors of circles and polygons). Example line segments are shown within the figures.

Although example line segments are shown drawn within the figures in Figure 1.1.18, recall that in order to be convex, all line segments need to be within the set, not just an example line segment.

However, the interior of the objects in Figure 1.1.19 are not. A line is shown in blue with point endpoints that are in the set, but the are parts of the line that are not in the set.

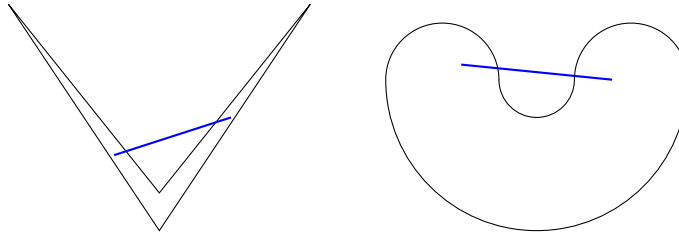


Figure 1.1.19 The interior of these shapes are not convex. Example line segments are drawn which have points that are outside the shape (sets).

The next lemma states the seemingly obvious fact that splitting the plane with a linear inequality results in a convex set.

Lemma 1.1.20 Let $S = \{(x, y) \mid ax + by \leq c\}$ with $a, b, c \in \mathbb{R}$ and $ab \neq 0$. The set S is convex.

Proof. Let (x_1, y_1) and (x_2, y_2) be two points in S . The line segment between them is the parametric equation

$$x(t) = x_1t + (1 - t)x_2 \quad y(t) = y_1t + (1 - t)y_2.$$

To show convexity, the line segment must be in S . That is,

$$\begin{aligned} a(x_1t + (1 - t)x_2) + b(y_1t + (1 - t)y_2) &= t(ax_1 + by_1) + (1 - t)(ax_2 + by_2) \\ &\leq tc + (1 - t)c \\ &= tc + c - ct = c \end{aligned}$$

And since the both points (x_1, y_1) and (x_2, y_2) are in S , then $a(x_1t + (1 - t)x_2) + b(y_1t + (1 - t)y_2) \leq c$, the line segment is in S , therefore the set is convex. ■

This showed that the linear inequality in $\mathbb{R}^2$ is convex, however this is true in any dimension.

Lemma 1.1.21 Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $S = \{\mathbf{x} \mid a_1x_1 + a_2x_2 + \dots + a_nx_n \leq c\}$ for $a_1, a_2, \dots, a_n, c$ constants. The set S is convex.

This lemma is the first step in showing the general convexity of more complex sets. The next one will give the next step.

Lemma 1.1.22 Let S and T be two convex sets. The intersection $S \cap T$ is convex.

The proof is left to the reader.

1.2 Using Software for Plotting

Objectives

- Using Desmos or other software to plot lines and linear inequalities.

1.2.1 Plotting Lines

Software can help with the plotting of lines, planes and linear inequalities. In [Subsection 1.1.1](#), the equation $3x + 4y = 12$ and linearly inequality $3x + 4y \leq 12$ was plotted.

Visit [Desmos](#) which will look like:

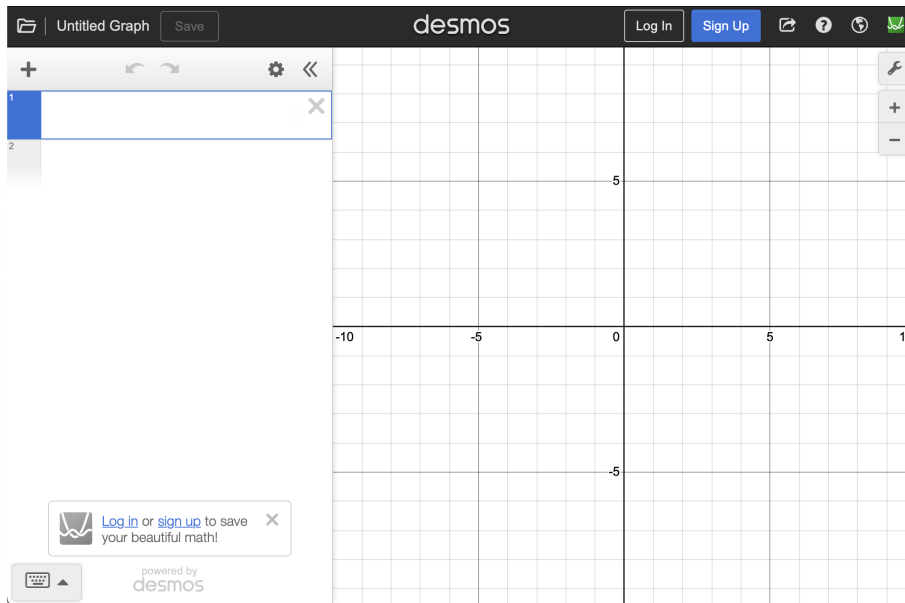
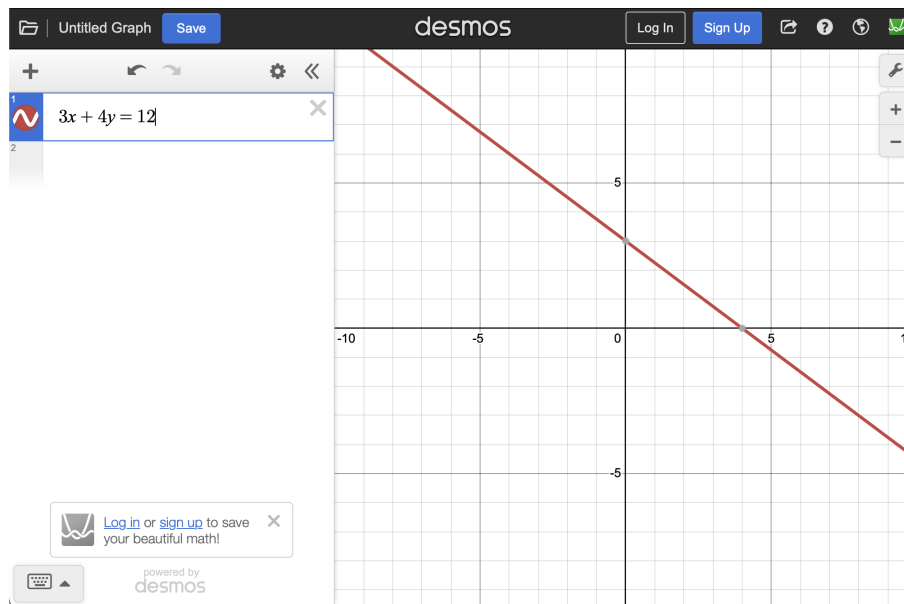
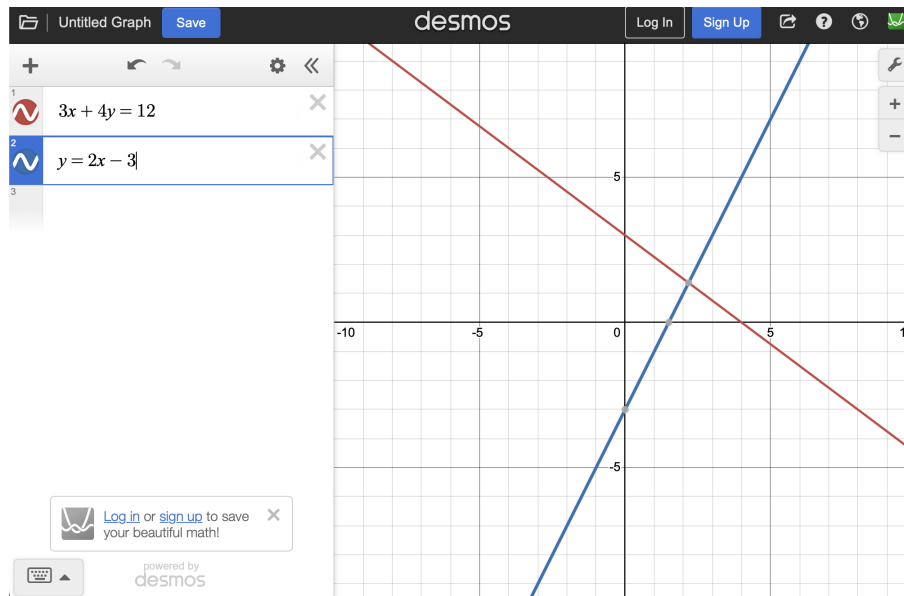


Figure 1.2.1

Enter $3x+4y=12$ in the first box (outlined in blue) and you should see

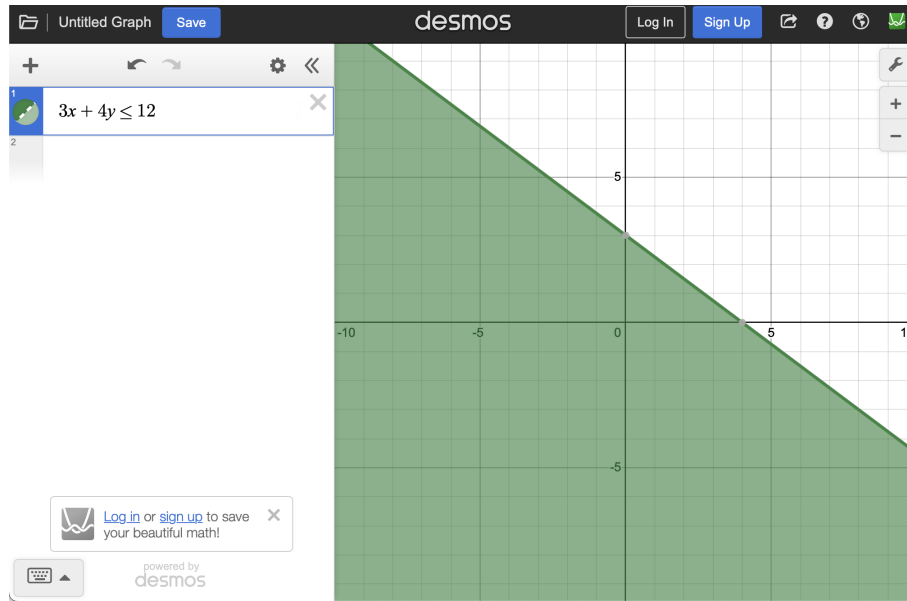
**Figure 1.2.2**

Additional lines can be plotted by adding them to the box below the previous line. For example, enter $y = 2x - 3$ results in

**Figure 1.2.3**

1.2.2 Plotting Inequalities and Feasible Sets

Inequalities can also be plotted. Return to desmos and remove all equations by hitting the X on the right side of each entry box. Now add $3x + 4y \leq 12$ where Desmos will automatically convert $\leq$ to $\leq$. The resulting plot will be

**Figure 1.2.4**

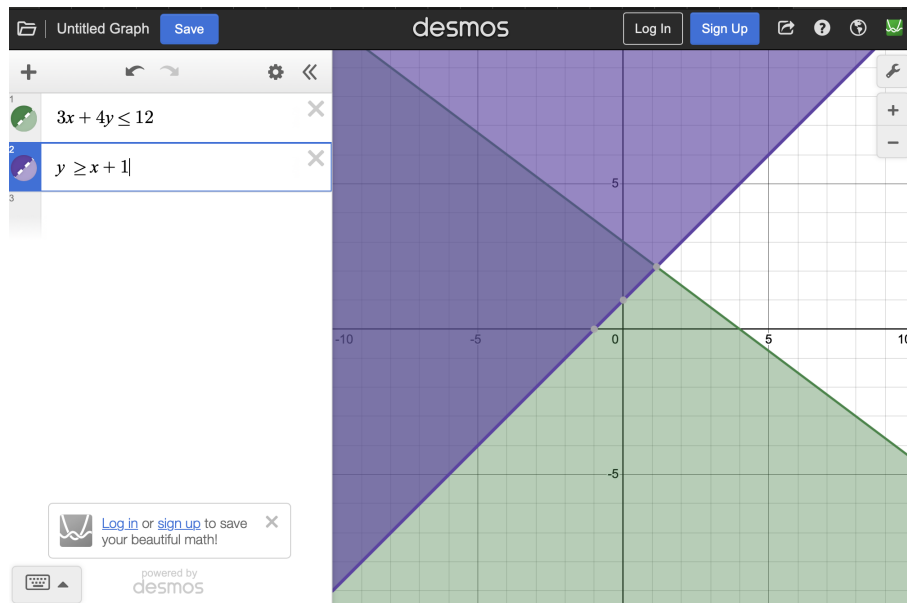
Note that the colors of all the plots may be different. Desmos usually cycles through the colors in a specific manner, but this may differ as you enter equations.

The graph shows both the line as solid and the color as a light green indicating that the region below and left of the line is part of the inequality.

This idea can be extended to plotting sets of inequalities to arrive at a feasible set. Recall that in [Section 1.1](#) the set of inequalities:

$$\begin{aligned} 3x + 4y &\leq 12 \\ y &\geq x + 1 \end{aligned}$$

was plotted. If we add the second inequality to Demos we get

**Figure 1.2.5**

With only two inequalities, visually the overlap (the set of points that satisfy both inequalities) is reasonably easy to see, however, as these grow, it can be more difficult to see. One way is to plot the opposite inequality and then the area of the plane with no coloring would be the resultant set.

Example 1.2.6 Use Desmos to plot the region that satisfies the following:

$$x - 2y \leq -2,$$

$$5x - 2y \geq 15,$$

$$3x + 5y \geq 75.$$

Solution. We open up Desmos and add all three of the inequalities, except switching each inequality. The result is

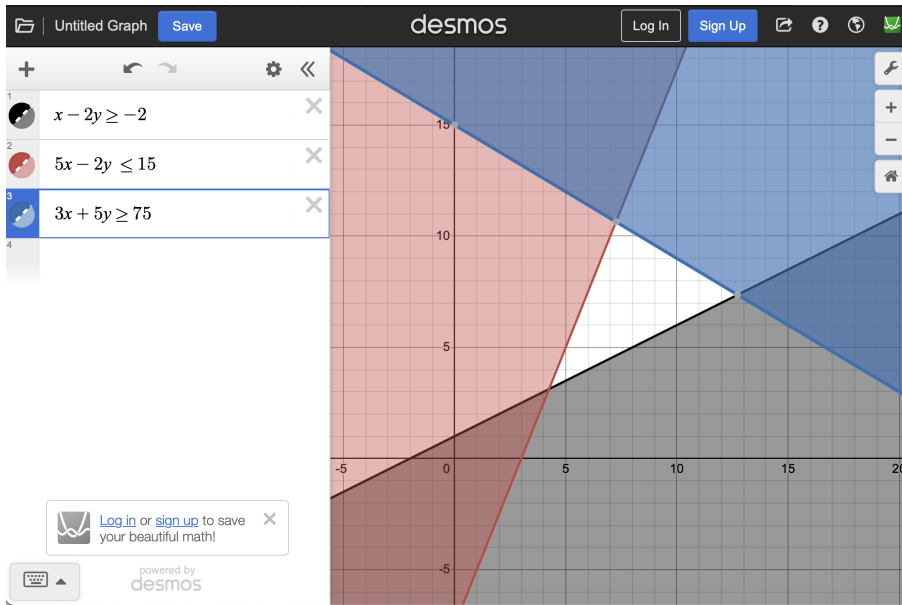


Figure 1.2.7

The set of points that satisfy the three inequalities is the triangular region without any shading. $\square$

1.2.3 Graphing Planes

As we saw in [Section 1.1](#), planes are another example of linear functions that we will encounter in this text. Desmos also has a [3D graphing calculator](#). Open up the webpage and it will look like the regular graphing calculator, however the default axes are in $\mathbb{R}^3$.

Add $x+2y+3z=12$ to the plot and you should get a plane. Spin the axes around and zoom in and out with the + and - buttons. You may get something that looks like

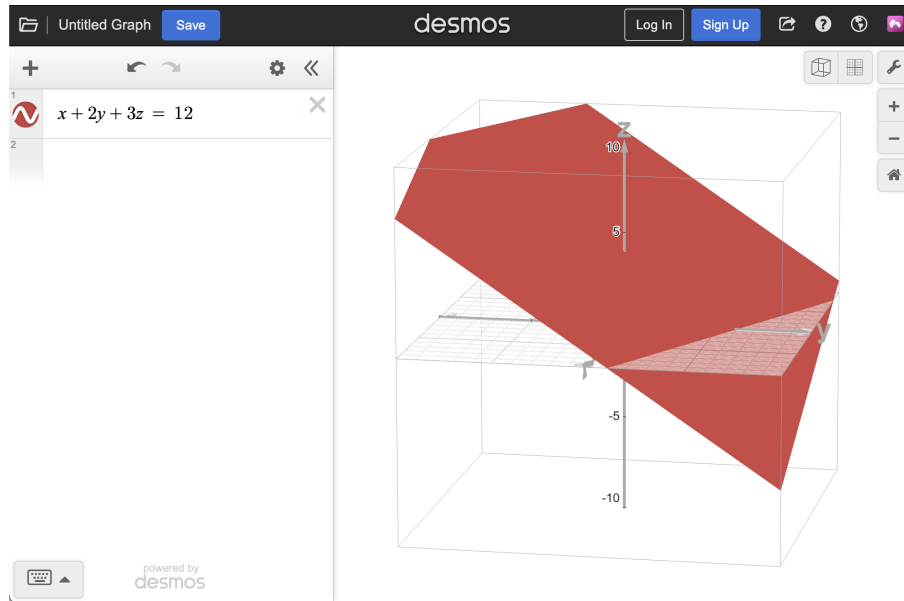


Figure 1.2.8

Inequalities can also be plotted, however, they are difficult to read. For example, change the $=$ in the previous plane to a $\leq$ or $\geq$ and you will see that half of the $\mathbb{R}^3$ space is now colored red (in my case). However, adding another inequality will be quite difficult to see the intersecting region.

1.3 Matrices and Matrix Reduction

Matrices are generally presented in either a linear algebra or matrix methods class. The material presented here is some basics behind matrices that are needed for performing linear optimization. This should be review for most readers of the text, but is helpful for the context and notation later in the text.

1.3.1 Linear Systems

Let's start with an example.

Example 1.3.1 Traffic Flow. A simple model of traffic flow can be represented by the following graph:

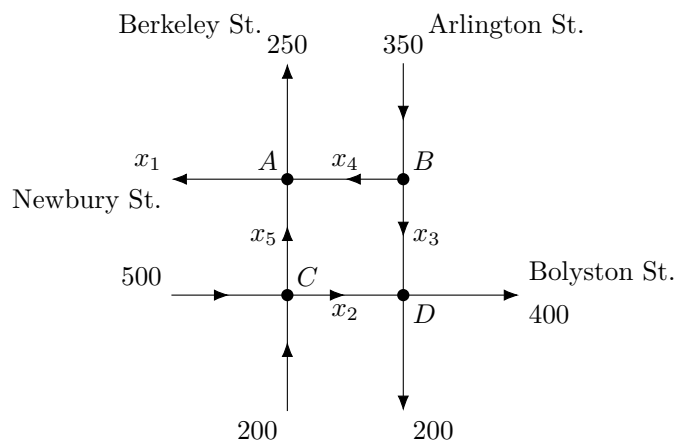


Figure 1.3.2 A map of a few streets in Boston where the arrows denote the direction of traffic flow (all of these streets are one-way) and the numbers represent the numbers of cars driving down the street in a given time period. The letters A through D will be the names of the intersections.

Write down the equations that balance each of the numbers of cars entering and leaving each of the intersections A, B, C and D .

Solution. In this case, we have:

$$x_4 + x_5 = x_1 + 250 \quad 350 = x_3 + x_4 \quad 700 = x_2 + x_5 \quad x_3 + x_2 = 600 \quad (1.3.1)$$

for intersections A, B, C and D respectively. $\square$

Definition 1.3.3

- A **linear combination** of $x_1, x_2, x_3, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n,$$

where the constants $a_1, a_2, \dots, a_n \in \mathbb{R}$ are the combinations coefficients.

- A **linear equation** has the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b \quad (1.3.2)$$

where $b \in \mathbb{R}$ is a constant and $a_1, a_2, \dots, a_n \in \mathbb{R}$. These are the lines, planes and hyperplanes discussed in [Section 1.1](#).

- The n -tuple $(s_1, s_2, \dots, s_n)$ **satisfies** or is a **solution** of [\(1.3.2\)](#) if this point satisfies [\(1.3.2\)](#) or

$$a_1s_1 + a_2s_2 + \dots + a_ns_n = b$$

- A **system of linear equations** or **linear system** is a set of linear equations:

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n &= b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n &= b_2, \\ &\vdots \\ &= \vdots \end{aligned} \quad (1.3.3)$$

$$a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m,$$

and this linear system has m equations and n unknowns (variables). The equations in (1.3.1) is an example of a linear system.

- The n -tuple $(s_1, s_2, \dots, s_n)$ **satisfies** or is a **solution** of (1.3.3) if this point satisfies *every* equation of (1.3.3).

◇

Example 1.3.4 Linear Equations. The following are linear equations:

$$\begin{aligned} 2x + 3y &= 6, & y_1 - y_2 + y_3 - y_4 &= 10, \\ 10x_1 - x_3 + 5x_5 &= 9, & \sum_{i=1}^{10} ix_i &= 0 \end{aligned}$$

where summation notation has been used in the last one and note the variable names can vary. The following equations are not linear:

$$\begin{aligned} x_1^2 + x_2 &= 6, & y_1y_2 + y_3 &= 5, \\ \frac{x_1 + x_2}{x_3} &= 6, & \sin(x + y) &= z \end{aligned}$$

Each of the equations in the latter group have multiplications, squares, division or other functions between variables. □

The next two examples give a way to determine if a point or n -tuple is a solution to a linear system.

Example 1.3.5 Showing a Point is a Solution to a Linear System.

Show that the point $(2, 3)$ is a solution of the linear system:

$$\begin{aligned} 3x_1 - x_2 &= 3 \\ 2x_1 + 4x_2 &= 16 \end{aligned}$$

Solution. Substitute $x_1 = 2$ and $x_2 = 3$ into both equations and check.

$$\begin{aligned} 3(2) - 3 &= 3, \\ 2(2) + 4(3) &= 16. \end{aligned}$$

Since each equation is satisfied at the point $(2, 3)$ is a solution to the linear system. □

Example 1.3.6 Show that $(200, 500, 100, 250, 200)$ and $(200, 450, 150, 200, 250)$ are both solutions to the linear system in (1.3.1).

Solution. In each case, the values must satisfy each equation. In the first case:

$$\begin{aligned} 250 + 200 &= 200 + 250, \\ 350 &= 100 + 250, \\ 700 &= 500 + 200, \\ 100 + 500 &= 600, \end{aligned}$$

and since equation equation is satisfied, it is a solution. In the second case:

$$200 + 250 = 200 + 250$$

$$350 = 150 + 200$$

$$700 = 450 + 250$$

$$150 + 450 = 600$$

and again, each equation is satisfied.

This shows that the linear system in (1.3.1) has more than one solution. In this case, there are different traffic levels that satisfy the equations. $\square$

1.3.2 Elementary Row Operations

We now seek a method to solve the above linear systems (and any linear system). Gauss' method and variations of it are the standard ways that all large linear systems are solved presently. The next section shows how to systematically solve a linear system using rules of algebra that keep linear equations linear.

Consider the following example of two linear equations:

$$x + 2y + 3z = 6 \tag{1.3.4}$$

$$-2x + 3y - z = 4 \tag{1.3.5}$$

We seek to find operations on these equations that keep these equations linear and don't change the solution. First, the order that we wrote them down doesn't matter and we could swap these equations as in:

$$-2x + 3y - z = 4$$

$$x + 2y + 3z = 6$$

The second possible operation is to multiply a single equation by a constant. For example, if we multiply (1.3.4) by 2, we get the system:

$$2x + 4y + 6z = 12$$

$$-2x + 3y - z = 4$$

The third operation is that we can add two equations and then replace one of the equations with the sum. For example adding (1.3.4) and (1.3.5) and putting the sum in the 2nd row results in

$$x + 2y + 3z = 6$$

$$-x + 5y + 2z = 10$$

A more common operation is to combine the last two operations and that is to multiply an equation by a constant and add to another. If we multiply (1.3.4) by 2 and add to (1.3.5) we get the system:

$$x + 2y + 3z = 6$$

$$7y + 5z = 16$$

Definition 1.3.7 Elementary Row Operations. The three operations

1. Two equations of the linear system are swapped,
2. An equation is multiplied by a nonzero number,
3. An equation is replaced by the sum of itself and a multiple of another equation,

are called the **elementary row operations** of a linear system. $\diamond$

1.3.2.1 Gauss's Method

The following example shows how to use the elementary row operations seen above to solve a linear system with a technique called *Gauss's Method*.

$$\begin{aligned}x - 2y + 3z &= 6, \\2x + 2y &= 6, \\-3x &+ z = 0.\end{aligned}$$

If one equation had only one variable, it would be quite easy to find the solution to that equation. The following steps will result in such a system. For example, if the first equation is multiplied by -2 and added to the second equation and the second equation is replaced (which we denote with $-2R_1 + R_2 \rightarrow R_2$), then the above equations are replaced with

$$\begin{aligned}-2R_1 + R_2 \rightarrow R_2 \quad \quad \quad & \begin{aligned}x - 2y + 3z &= 6, \\6y - 6z &= -6, \\-3x &+ z = 0.\end{aligned}\end{aligned}$$

Next, we'd like to get rid of the x term in the 3rd equation. We can do that by multiplying the first equation by 3 and adding to the 3rd and writing down this operation as well we get:

$$\begin{aligned}3R_1 + R_3 \rightarrow R_3 \quad \quad \quad & \begin{aligned}x - 2y + 3z &= 6, \\6y - 6z &= -6, \\-6y + 10z &= 18.\end{aligned}\end{aligned}$$

Next we will eliminate the y term in the 3rd equation. We can do that by adding equations 2 and 3 and putting the result in equation 3.

$$\begin{aligned}R_2 + R_3 \rightarrow R_3 \quad \quad \quad & \begin{aligned}x - 2y + 3z &= 6, \\6y - 6z &= -6 \\4z &= 12.\end{aligned}\end{aligned} \tag{1.3.6}$$

At this point we can solve for z in the third equation of (1.3.6) to get $z = 3$. From this, plug in $z = 3$ into the second equation of (1.3.6) to get

$$\begin{aligned}6y - 6(3) &= -6 \\6y &= 12\end{aligned}$$

$$y = 2$$

and finally we can use $z = 3$ and $y = 2$ to substitute into the first equation of (1.3.6) to get

$$x - 2(2) + 3(3) = 6$$

$$x - 4 + 9 = 6$$

$$x = 1$$

therefore, $x = 1$. The point $(1, 2, 3)$ is a solution to linear system. Because this is the only point that satisfies all equations, this point is *unique*. We will also use the terminology triple to describe $(1, 2, 3)$.

Theorem 1.3.8 Gauss's Method. *If a linear system (A) is changed into a second linear system (B) by one of the elementary row operations then linear systems (A) and (B) have the same set of solutions.*

Proof. We will consider only the first operation in this proof. Let's assume that we swap equations i and j , thus system (A),

$$\begin{array}{rcl} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n & = & b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n & = & b_2, \\ & \vdots & \\ a_{i,1}x_1 + a_{i,2}x_2 + \cdots + a_{i,n}x_n & = & b_i, \\ & \vdots & \\ a_{j,1}x_1 + a_{j,2}x_2 + \cdots + a_{j,n}x_n & = & b_j, \\ & \vdots & \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n & = & b_m, \end{array}$$

is transformed to

$$\begin{array}{rcl} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n & = & b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n & = & b_2, \\ & \vdots & \\ a_{j,1}x_1 + a_{j,2}x_2 + \cdots + a_{j,n}x_n & = & b_j, \\ & \vdots & \\ a_{i,1}x_1 + a_{i,2}x_2 + \cdots + a_{i,n}x_n & = & b_i, \\ & \vdots & \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n & = & b_m, \end{array}$$

Let $(s_1, s_2, \dots, s_n)$ be a solution of (A) if it exists and note it may be one of many n -tuples in the solution. Thus it satisfies each equation of linear system (A). Since the exact same equations are in (B) in just a different order, $(s_1, s_2, \dots, s_n)$ is a solution to (B). If there is more than one n -tuple in the solution to (A), repeat this for every one. If there is no solution to (A), then there will be no solution to (B) since it is the same set of equations.

Proof of #2 and #3 above are quite similar and are not shown. ■

Note 1.3.9 Although Gauss' Method is very flexible, generally, one tries to eliminate all x_1 terms in all equations below equation 1, x_2 terms in all equations below equation 2 and so on.

1.3.3 Matrices

When we solved a linear system above, you probably noticed that the variables didn't play much of a role, the coefficients changed and the variables stayed in the same place as we solved for them. Because of this, we will adopt that technique to only work with the coefficients. We first define a matrix that we will use to solve linear systems for the rest of this book.

Definition 1.3.10 An m by n **matrix** is a rectangular grid of numbers with m rows and n columns. The **size** of a matrix is the pair of number m and n and is typically written m by n or $m \times n$. Each number of the matrix is called the **entry** or **element**. $\diamond$

Example 1.3.11 Matrices. The following are examples of matrices:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad \begin{bmatrix} \pi & 2 \\ -1 & 4 \\ 2.5 & 7/3 \\ -11 & 1000 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \\ 9 \end{bmatrix} \quad [8 \quad -3 \quad 2]$$

$\square$

A common notation for a matrix is to use upper case letters (and often bold), for example A , B and C are common matrices. The entry or element in the i th row and j th column of A is $a_{i,j}$. Also, the set of all m by n matrices is denoted $\mathcal{M}_{m \times n}$.

Definition 1.3.12 A matrix with only 1 row is called a **row vector**. A matrix with only 1 column is called a **column vector**. The **size** of a vector is the number of rows (for a column vector) or the number of columns (for a row vector). $\diamond$

The 3rd matrix in [Example 1.3.11](#) above is a column vector of size 5. The 4th matrix is a row vector of size 3. Vectors are very important and will see them throughout this book.

1.3.4 Matrices and Linear Systems

Let's look at the following linear system.

$$\begin{aligned} x + 2y &= 7, \\ 4x - 3y &= 6. \end{aligned}$$

We will write the coefficients of the variables and the right hand side in a matrix called an **augmented coefficient matrix**. The following is this matrix for the linear system above.

$$\left[\begin{array}{cc|c} 1 & 2 & 7 \\ 4 & -3 & 6 \end{array} \right]$$

and the two rows of the matrix represent the two equations. The first column represents the coefficient of the x variable, and the second column represents the coefficient of the y variable. The last column is the right hand

sides of the linear system, and the vertical line occurs where the equal sign is in the equations.

Example 1.3.13 Linear System as an Augmented Matrix. Write down the following linear system as an augmented matrix:

$$\begin{aligned} 3x - 2y + z &= 11, \\ 2x + 4y - 3z &= 2, \\ -4x + y + 4z &= 0. \end{aligned}$$

Solution.

$$\left[\begin{array}{ccc|c} 3 & -2 & 1 & 11 \\ 2 & 4 & -3 & 2 \\ -4 & 1 & 4 & 0 \end{array} \right]$$

□

And the next example shows how to rewrite a matrix as a linear system.

Example 1.3.14 Augmented Matrix as a Linear System. Write down the following augmented matrix as a linear system:

$$\left[\begin{array}{ccc|c} 3 & 0 & -1 & 7 \\ -2 & 1 & 5 & 6 \\ 0 & 3 & 4 & 8 \end{array} \right]$$

Solution. In this case, we are doing the opposite of the step above. That is, take a matrix and find the linear system.

Since the matrix has 4 columns, there are 3 variables (the last column is the right hand side). Let's use x , y and z .

$$\begin{aligned} 3x + 0y - z &= 7 \\ -2x + y + 5z &= 6 \\ 0x + 3y + 4z &= 8 \end{aligned}$$

Alternatively, we could have used x_1, x_2 and x_3 as variables.

□

1.3.5 Row Operations and Matrices

We can perform the same row operations that we have seen above on matrices.

Row Swap	$R_i \leftrightarrow R_j$ swaps rows i and j .
Multiply by a number	cR_i multiplies row i by c . The notation $cR_i \rightarrow R_i$ is also used.
Multiply by a number and add to another	$cR_i + R_j$ multiplies row i by c and adds to row j . The notation $cR_i + R_j \rightarrow R_j$ is also used.

In the next section, we will formalize the solving of linear system using matrices, however, the last example here reproduces the same solution technique for the example in [Subsubsection 1.3.2.1](#).

Example 1.3.15 Solve the linear system

$$\begin{aligned}x - 2y + 3z &= 6, \\2x + 2y &= 6, \\-3x &+ z = 0.\end{aligned}$$

by first writing as an augmented coefficient matrix and then performing elementary row operations on the matrix.

Solution. First, the augmented coefficient matrix is

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 6 \\ 2 & 2 & 0 & 6 \\ -3 & 0 & 1 & 0 \end{array} \right]$$

Now perform the same elementary row operations as above. We first seek to change the second row, first column to a 0

$$-2R_1 + R_2 \rightarrow R_2 \quad \left[\begin{array}{ccc|c} 1 & -2 & 3 & 6 \\ 0 & 6 & -6 & -6 \\ -3 & 0 & 1 & 0 \end{array} \right]$$

and notice that this is the matrix representation of the second step in the solution listed in [Subsubsection 1.3.2.1](#). Next, we seek to change the 3rd row, first column from -3 to a 0.

$$3R_1 + R_3 \rightarrow R_3 \quad \left[\begin{array}{ccc|c} 1 & -2 & 3 & 6 \\ 0 & 6 & -6 & -6 \\ 0 & -6 & 10 & 18 \end{array} \right]$$

And since we seek to get only a single variable in the third row, only one more step is needed.

$$R_2 + R_3 \rightarrow R_3 \quad \left[\begin{array}{ccc|c} 1 & -2 & 3 & 6 \\ 0 & 6 & -6 & -6 \\ 0 & 0 & 4 & 12 \end{array} \right]$$

At this point, if we return the augmented matrix to a set of equations, we get

$$\begin{aligned}x - 2y + 3z &= 6 \\6y - 6z &= -6 \\4z &= 12\end{aligned}$$

and we can get the same solution using the same steps as above to get $z = 3, y = 2, x = 1$. $\square$

We will see in the next section that actually if we continue working with the matrix, we can more easily find the solution as well as find solutions for a broader range of linear systems.

Example 1.3.16 Solve

$$\left[\begin{array}{ccc|c} -4 & 1 & -2 & 5 \\ 3 & -1 & 0 & 2 \\ 1 & 0 & -3 & -3 \end{array} \right] \quad (1.3.7)$$

using Gauss-Jordan Elimination

Solution. If this is done "by hand", one would probably swap rows 1 and 4 to get a 1 in the upper left, however let's skip and get a 1 there by dividing through by -4 , then eliminate the rest of the column:

$$\begin{array}{l} -\frac{1}{4}R_1 \rightarrow R_1, \\ -3R_1 + R_2 \rightarrow R_2, \\ -R_1 + R_3 \rightarrow R_3 \end{array} \quad \left[\begin{array}{ccc|c} 1 & -1/4 & 1/2 & -5/4 \\ 0 & -1/4 & -3/2 & 23/4 \\ 0 & 1/4 & -7/2 & -7/4 \end{array} \right]$$

Typically we now want to get a 1 in the 2nd row, 2nd column of the matrix and can be done by multiplying row 2 by -4 and then eliminating the rest of the column with

$$\begin{array}{l} -4R_2 \rightarrow R_2, \\ \frac{1}{4}R_2 + R_1 \rightarrow R_1, \\ -\frac{1}{4}R_2 + R_3 \rightarrow R_3, \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & -7 \\ 0 & 1 & 6 & -23 \\ 0 & 0 & -5 & 4 \end{array} \right]$$

Lastly, we will make row 3, column 3 a 1 and then use this to eliminate the rest of the 3rd column with

$$\begin{array}{l} -\frac{1}{5}R_3 \rightarrow R_3, \\ -2R_3 + R_1 \rightarrow R_1, \\ -6R_3 + R_2 \rightarrow R_2, \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & -27/5 \\ 0 & 1 & 0 & -91/5 \\ 0 & 0 & 1 & -4/5 \end{array} \right] \quad (1.3.8)$$

And this reveals the solution is the vector $[-27/5 \quad -91/5 \quad -4/5]^\top$ $\square$

Checkpoint 1.3.17 Put the linear system from (1.3.1) in a augmented coefficient matrix and use row operations to put the matrix in reduced row-echelon form.

Solution. First, ensure that all of the variables are on the left of the equations in (1.3.1) and the constants on the right. This can be written as the augmented coefficient matrix,

$$\left[\begin{array}{cccc|c} -1 & 0 & 0 & 1 & 1 & 250 \\ 0 & 0 & 1 & 1 & 0 & 350 \\ 0 & 1 & 0 & 0 & 1 & 700 \\ 0 & 1 & 1 & 0 & 0 & 600 \end{array} \right]$$

Row operations are now performed

$$\begin{array}{l} -R_1 \rightarrow R_1 \\ \\ R_2 \leftrightarrow R_3 \\ \\ -R_2 + R_4 \rightarrow R_4 \end{array} \quad \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 & -250 \\ 0 & 0 & 1 & 1 & 0 & 350 \\ 0 & 1 & 0 & 0 & 1 & 700 \\ 0 & 1 & 1 & 0 & 0 & 600 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 & -250 \\ 0 & 1 & 0 & 0 & 1 & 700 \\ 0 & 0 & 1 & 1 & 0 & 350 \\ 0 & 1 & 1 & 0 & 0 & 600 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 & -250 \\ 0 & 1 & 0 & 0 & 1 & 700 \\ 0 & 0 & 1 & 1 & 0 & 350 \\ 0 & 0 & 1 & 0 & -1 & -100 \end{array} \right]$$

$$\begin{array}{l}
-R_3 + R_4 \rightarrow R_4 \\
\\
-R_4 \rightarrow R_4 \\
\\
\begin{array}{l}
R_4 + R_1 \rightarrow R_1, \\
-R_4 + R_3 \rightarrow R_3,
\end{array}
\end{array}
\left[\begin{array}{ccccc|c}
1 & 0 & 0 & -1 & -1 & -250 \\
0 & 1 & 0 & 0 & 1 & 700 \\
0 & 0 & 1 & 1 & 0 & 350 \\
0 & 0 & 0 & -1 & -1 & -450 \\
\\
1 & 0 & 0 & -1 & -1 & -250 \\
0 & 1 & 0 & 0 & 1 & 700 \\
0 & 0 & 1 & 1 & 0 & 350 \\
0 & 0 & 0 & 1 & 1 & 450 \\
\\
1 & 0 & 0 & 0 & 0 & 200 \\
0 & 1 & 0 & 0 & 1 & 700 \\
0 & 0 & 1 & 0 & -1 & -100 \\
0 & 0 & 0 & 1 & 1 & 450
\end{array} \right]$$

Remark 1.3.18 A fourth operation. Another handy row operation is that of replacing a row with linear combination of itself and another row. In general this is

$$c_i R_i + c_j R_j \rightarrow R_j$$

Since this is a combination of two row operations together, we will use it, however, not consider it a separate (fourth) row operation.

1.3.6 Matrix Pivots

The technique of getting a 1 somewhere in a column and 0's elsewhere is known as performing a **matrix pivot**. In fact, this is what was done in [Example 1.3.16](#) and is the standard step in performing Gauss-Jordan Elimination in the goal to get a matrix in reduced row-echelon form.

If we desire a 1 at row i in column j , then the following row operations will produce this 1 and zeros throughout the remainder of the column. Let the matrix have coefficients $a_{i,j}$.

1. The step

$$\frac{1}{a_{i,j}} R_i \rightarrow R_i$$

will get a 1 at row i , column j .

2. The step

$$-a_{k,j} R_i + R_k \rightarrow R_k \quad \text{for } k \neq i$$

will zero out the rest of the column. This is called the **elimination step**.

If we look at [Example 1.3.16](#), then these steps succeeded in producing the desired structure, however, notice that it introduces fractions and always will if the pivot is not a 1. If we adapt the steps above, we can perform a pivot that keeps fractions at bay.¹

¹Realistic matrices will often have non integer entries, so this technique does not work in general, however it does make operations on integer matrices easier to do—and there are a number of realistic matrices contain only integers.

Algorithm 1.3.19 Fraction Free Pivot. *If the matrix A has only integer entries, then the following will perform a **fraction-free pivot** at row i , column j , which will eliminate a column, but not produce a 1 at the pivot point. The result will be a matrix with all integer entries.*

Let $a_{i,j}$ be the coefficient of a $m \times n$ matrix (and an integer). If $a_{i,j} < 0$, then perform:

$$-R_i \rightarrow R_i.$$

Then the following row operations

$$-a_{k,j}R_i + a_{i,j}R_k \rightarrow R_k \quad \text{for } k = 1, \dots, m, k \neq i$$

will eliminate column j .

Lastly, let d be the positive integer that is the (i, j) entry. The other rows will be multiples of d and the row operation

$$\frac{1}{d}R_k \rightarrow R_k \quad \text{for } i \neq k$$

and any other pivots will now have a entry of d .

The following example applies this algorithm to a previously solved linear system.

Example 1.3.20 Solve (1.3.7) using Algorithm 1.3.19.

Solution. The first step use the matrix pivot in Algorithm 1.3.19 to perform a matrix pivot on row 1, column 1:

$$\begin{array}{l} -3R_1 - 4R_2 \rightarrow R_2 \\ -R_1 - 4R_3 \rightarrow R_3 \\ -R_1 \rightarrow R_1 \end{array} \quad \left[\begin{array}{ccc|c} 4 & -1 & 2 & -5 \\ 0 & 1 & 6 & -23 \\ 0 & -1 & 14 & 7 \end{array} \right]$$

Next, let's perform a pivot about row 2, column 2.

$$\begin{array}{l} R_2 + R_1 \rightarrow R_1 \\ R_2 + R_3 \rightarrow R_3 \end{array} \quad \left[\begin{array}{ccc|c} 4 & 0 & 8 & -28 \\ 0 & 1 & 6 & -23 \\ 0 & 0 & -20 & -16 \end{array} \right]$$

The first and third rows are both multiples of 4, the previous pivot. If we perform:

$$\begin{array}{l} \frac{1}{4}R_1 \rightarrow R_1, \\ \frac{1}{4}R_3 \rightarrow R_3 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & -7 \\ 0 & 1 & 6 & -23 \\ 0 & 0 & 5 & -4 \end{array} \right]$$

Lastly, let's perform another FF pivot on row 3, column 3:

$$\begin{array}{l} -6R_3 + 5R_2 \rightarrow R_2 \\ -2R_3 + 5R_1 \rightarrow R_1 \end{array} \quad \left[\begin{array}{ccc|c} 5 & 0 & 0 & -27 \\ 0 & 5 & 0 & -91 \\ 0 & 0 & 5 & -4 \end{array} \right]$$

And compare this matrix to that in the final reduced-row echelon form of (1.3.8) which if each row of the matrix above is multiplied through by $1/5$, then (1.3.8) is created. The solution can be read from the matrix as the left hand side divided by d , in this case 5. This solution is

$$\mathbf{x} = \frac{1}{5} \begin{bmatrix} -27 & -91 & -4 \end{bmatrix}^\top$$

and note that this is the same solution as in Example 1.3.16 and in many ways is nicer in that all of the entries of the matrix are integers. $\square$

In this course, we will perform matrix pivots, but not in a way to put the matrix in Reduced Row Echelon Form. Instead, the matrices we use will have more variables than rows, resulting in free variables and depending on different conditions, we will use different free variables. Let's look at another example:

Example 1.3.21 Let the following matrix have variables $x_1, x_2, \dots, x_6$. Solve the matrix for x_1, x_3 and x_6 and have free variables x_2, x_4 and x_5 . Use the Fraction-Free pivots in the solve.

$$\left[\begin{array}{cccccc|c} 3 & 1 & 0 & 0 & 2 & -7 & 11 \\ -2 & 0 & 3 & 1 & -3 & 0 & 15 \\ 1 & 1 & 0 & 0 & 2 & -2 & 13 \end{array} \right]$$

Solution. If we are trying to solve for x_1 , then we can perform a pivot about an element in column 1. Let's pivot about row 1, column 1,

$$\begin{array}{l} -2R_1 - 3R_2 \rightarrow R_2, \\ R_1 - 3R_3 \rightarrow R_3 \end{array} \quad \left[\begin{array}{cccccc|c} 3 & 1 & 0 & 0 & 2 & -7 & 11 \\ 0 & -2 & -9 & -3 & 5 & 14 & -67 \\ 0 & -2 & 0 & 0 & -4 & -1 & -28 \end{array} \right]$$

Next, we are looking to solve for x_3 and since row 1 gives a solution for x_1 , we don't want to mess that up and instead will choose row 2 to perform a pivot, so pivot about row 2, column 3:

$$\begin{array}{l} 0R_2 + 9R_1 \rightarrow R_1 \\ 0R_2 + 9R_3 \rightarrow R_3 \end{array} \quad \left[\begin{array}{cccccc|c} 27 & 9 & 0 & 0 & 18 & -63 & 99 \\ 0 & -2 & -9 & -3 & 5 & 14 & -67 \\ 0 & -18 & 0 & 0 & -36 & -9 & -252 \end{array} \right]$$

Note that we didn't really need to do any steps here since the form was correctly. However, for consistency between steps, we did this anyway.

Now, we perform some cleanup. First, let's make the 2nd row, 2nd column positive. and then since the first and 3rd rows are multiples of 3, the previous value of d , we will multiply through by $1/3$.

$$\begin{array}{l} \frac{1}{3}R_1 \rightarrow R_1, \\ \frac{1}{3}R_3 \rightarrow R_3, \end{array} \quad \left[\begin{array}{cccccc|c} 9 & 3 & 0 & 0 & 6 & -21 & 33 \\ 0 & 2 & 9 & 3 & -5 & -14 & 67 \\ 0 & -6 & 0 & 0 & -12 & -3 & -84 \end{array} \right]$$

Lastly, we want to solve for x_6 , so we will pivot about row 3, column 6 with the following row operations

$$\begin{array}{l} -21R_3 + 3R_1 \rightarrow R_1, \\ -14R_3 + 3R_2 \rightarrow R_2 \end{array} \quad \left[\begin{array}{cccccc|c} 27 & 135 & 0 & 0 & 270 & 0 & 1863 \\ 0 & 90 & 27 & 9 & 153 & 0 & 1377 \\ 0 & -6 & 0 & 0 & -12 & -3 & -84 \end{array} \right]$$

And again with some cleanup to the pivot in row 3, column 6 to be +3 and the pivots at row 1, column 1 and row 2, column 2 to also be 3.

$$\begin{array}{l} -R_3 \rightarrow R_3 \\ -\frac{1}{9}R_2 \rightarrow R_2 \\ -\frac{1}{9}R_1 \rightarrow R_1 \end{array} \quad \left[\begin{array}{cccccc|c} 3 & 15 & 0 & 0 & 30 & 0 & 207 \\ 0 & 10 & 3 & 1 & 17 & 0 & 153 \\ 0 & 6 & 0 & 0 & 12 & 3 & 84 \end{array} \right]$$

We can actually write down the equation that this matrix represents as

$$\begin{aligned}3x_1 + 15x_2 + 30x_5 &= 207 \\10x_2 + 3x_3 + x_4 + 17x_5 &= 153 \\6x_2 + 12x_5 + 3x_6 &= 84\end{aligned}$$

and solve for x_1, x_3 and x_6 as

$$\begin{aligned}x_1 &= \frac{1}{3}(207 - 15x_2 - 30x_5) \\x_2 &= \frac{1}{3}(153 - 10x_2 - x_4 - 17x_5) \\x_6 &= \frac{1}{3}(84 - 6x_2 + 2x_4 - 3x_5)\end{aligned}$$

And note that the three variables on the right hand side are free variables. This example will be similar to matrices in this class and understanding this example will get you far in understanding the matrices using here. $\square$

As we will see in this class, the matrices presented here are quite small, and clearly take a lot of operations to perform and the slightly arithmetic error results in the wrong answer. Once you have the idea under your belt, you should move onto using software to do these operations and we will show how to do this in the next section.

1.3.7 Geometry of Linear Systems of 2 variables

If we consider only linear systems with two variables, then each equation is just a line, which we can graph on a set of axes. We are going to examine the geometry of linear systems of two variables.

First, we are going to look at three linear systems that each have different types of solutions. We can see the solutions by looking at the graphs.

- The graph of each line in the linear system

$$\begin{aligned}x + 2y &= 5, \\2x - 3y &= -4,\end{aligned}\tag{1.3.9}$$

is

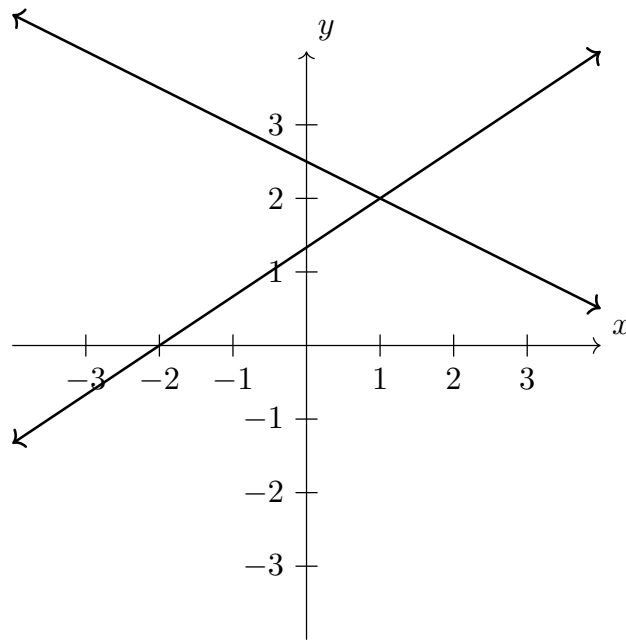


Figure 1.3.22 Plot of two intersecting lines

The solution of the system is the point at which they cross. In this case it looks like the point $(1, 2)$ or $x = 1$ and $y = 2$. This is an example where there is only one solution (or a **unique solution**).

- If we graph the lines in the linear system

$$\begin{aligned} 3x + 7y &= 10, \\ 6x + 14y &= 5, \end{aligned} \tag{1.3.10}$$

we get

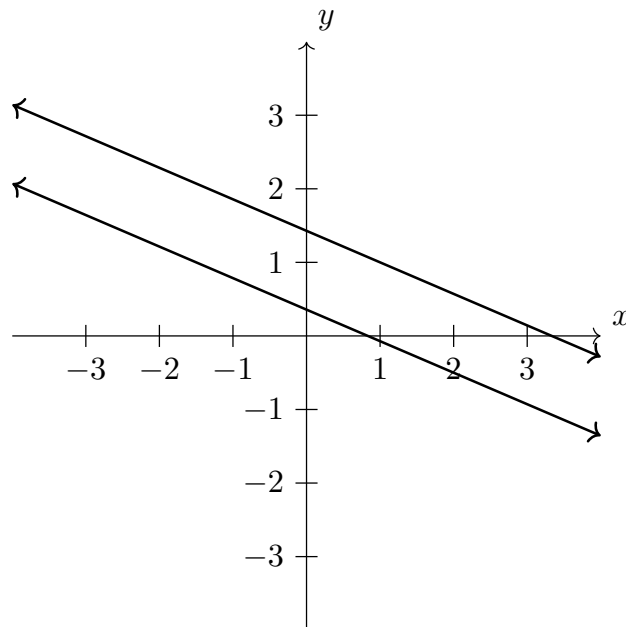


Figure 1.3.23 Plot of two parallel lines

As you can see, it doesn't appear that the lines cross anywhere. In fact, they don't because the lines are parallel. This is an example of a linear system with **no solution**.

- The linear system

$$\begin{aligned}x + 2y &= 5, \\ 2x + 4y &= 10,\end{aligned}\tag{1.3.11}$$

has the following graph for each line

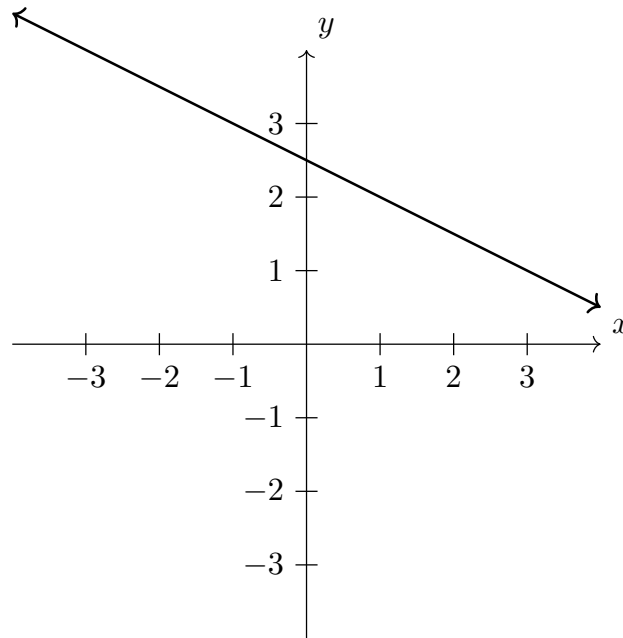


Figure 1.3.24 A plot of two equivalent lines

It appears that there is only one line. This is because both lines have the same graph. Each point on the line is a solution to the linear system and since there are an infinite number of such points, this is an example of a linear system with **infinite number of solutions**.

You can see if you have two lines, each with two variables, the example in (1.3.9) is what happens if the two slopes are different. In this case, there is one (or a **unique**) solution.

In the other two cases as in equations (1.3.10) and (1.3.11), both sets of lines have the same slope. In the case of (1.3.10) the lines have different y -intercepts, and therefore the lines are parallel and thus there is **no solution** to the linear system. In the case of (1.3.11), both the slope and y -intercepts are equal, so the lines are equal and thus any point on the line is a solution, and this system has an **infinite number of solutions**.

1.4 Using Software to Perform Matrix Pivots

Objectives

- Using WebCAS to perform row operations and pivots on matrices.

In the previous section, we performed row operations and matrix pivots to develop solutions to linear systems. In this section, we will show software to do this.

Almost all software that handles matrices have some way to do row operations on the matrix. However, often it is complicated to do. Clearly any software that will help with the understanding here should be used. We will show using a website called [WebCAS](#) that handles matrices in a very nice way. In this section, we will show some of the features.

1.4.1 Using WebCAS to do Row Operations

In this section, we will enter matrices into WebCAS and perform row operations on the matrix. Let's start with the matrix that we saw in [Example 1.3.21](#). From the WebCAS main page, select the *Gaussian Eliminator* tool. Within the large box, enter the matrix as numbers separated by one or more spaces and each row on a separate line. This matrix can go in as

Enter a Matrix

```
3 1 0 0 2 -7 11
-2 0 3 1 -3 0 15
1 1 0 0 2 -2 13
```

Clear Matrix Enter Matrix

Figure 1.4.1 Screenshot of WebCAS, a web-based software to do matrix operations.

and then click *Enter Matrix* and you should see the matrix displayed in a mathematical form.

Note 1.4.2 Although the matrix above probably does not have the vertical line on the last column. This is a matrix decoration—it's purpose is to separate out the last column visually. To get this, click the gear icon in the top bar and the first option is *Vertical Line Mode*. Select *Before Last Column* then click *Save Changes*.

You won't see them immediately, but if you click *Restart* and then enter the matrix again, you should see the results that you expect.

$$\left[\begin{array}{cccccc|c} 3 & 1 & 0 & 0 & 2 & -7 & 11 \\ -2 & 0 & 3 & 1 & -3 & 0 & 15 \\ 1 & 1 & 0 & 0 & 2 & -2 & 13 \end{array} \right]$$

Figure 1.4.3 The mathematical output of the matrix above put into WebCAS.

To perform row operations in the Gaussian Eliminator, the following are supported

Multiply Row by a Constant	If we want to multiply a row by a constant, like $3R_2 \rightarrow R_2$, then enter 3R2 -> R2
-----------------------------------	--

Multiply a Row by a Constant and add to another row	Mathematically, if you want to do $R_1 + 3R_2 \rightarrow R_2$, in WebCAS, enter <code>R1+3R2 -> R2</code>
Linear Combination of Rows	A linear combination of rows is the above two operations above mushed together. This is often not an elementary row operation because for certain operations (like finding a determinant), one needs to be careful. However, this is a nice convenient operation for matrix pivots. If you want to do $-2R_1 + 3R_2 \rightarrow R_2$, then enter <code>-2R1+3R2 -> R2</code>
Row swaps	The row swap $R_1 \leftrightarrow R_2$ can be done with <code>R1 <-> R2</code> .

For the operations except the Row Swap, if you leave off the `-> R2` or an row, it will use the previous row as the row to insert. This way, complicated row operations can be done.

Returning to the matrix above from [Example 1.3.21](#), we wanted to perform a matrix pivot about row 1, column 1. This was done with the two operations $-2R_1 - 3R_2 \rightarrow R_2$, $R_1 - 3R_3 \rightarrow R_3$ and in the Gaussian Eliminator, you can add both operations in the same box with `-2R1-3R2->R2, R1-3R3->R3`. (Or drop the arrow and last row). Hitting the *Enter* button (or the *enter* key on your keyboard), you should see:

$$\begin{array}{l} -2R_1 - 3R_2 \rightarrow R_2 \\ R_1 - 3R_3 \rightarrow R_3 \end{array} \quad \left[\begin{array}{cccccc|c} 3 & 1 & 0 & 0 & 2 & -7 & 11 \\ 0 & -2 & -9 & -3 & 5 & 14 & -67 \\ 0 & -2 & 0 & 0 & -4 & -1 & -28 \end{array} \right]$$

Figure 1.4.4 Row operations on the above matrix performed in WebCAS. This is a matrix pivot about row 1, column 1 and the image is the result of the two operations.

Note 1.4.5 Recall that on a matrix pivot the goal is to zero out the column that you are pivoting about (and we will want to get a positive number on the pivot element). Therefore, it should be clear that the pivot was done correctly. If not, there is an *Undo* button which will bring back the previous matrix and reinserts the row operation. Edits can be done and re-entered.

From my experience over many years, this is the students' favorite feature of this tool.

Continuing to do a matrix pivot on row 2, column 3, we do the following operation:

$$\begin{array}{l} 0R_2 + 9R_1 \rightarrow R_1 \\ 0R_2 + 9R_3 \rightarrow R_3 \end{array} \quad \left[\begin{array}{cccccc|c} 27 & 9 & 0 & 0 & 18 & -63 & 99 \\ 0 & -2 & -9 & -3 & 5 & 14 & -67 \\ 0 & -18 & 0 & 0 & -36 & -9 & -252 \end{array} \right]$$

Figure 1.4.6 Row operations on the above matrix performed in WebCAS. This is a matrix pivot about row 2, column 3 and the image is the result of the two operations.

Note that as we described in [Example 1.3.21](#), this operation wasn't quite needed (the second row already is solved for x_3). However, we did it to fit in with the standard pivot operations.

To finish the pivot, we need to negative the second row and then multiply the other two rows by $1/3$, where 3 is used as the pivot value of the previous

step.

$$\begin{array}{l} -1R_2 \rightarrow R_2 \\ \frac{1}{3}R_1 \rightarrow R_1 \\ \frac{1}{3}R_3 \rightarrow R_3 \end{array} \quad \left[\begin{array}{cccccc|c} 9 & 3 & 0 & 0 & 6 & -21 & 33 \\ 0 & 2 & 9 & 3 & -5 & -14 & 67 \\ 0 & -6 & 0 & 0 & -12 & -3 & -84 \end{array} \right]$$

Figure 1.4.7 Continued work on the previous pivot. The second row is negated to ensure that the pivot element is positive. Then the other two rows are multiplied by $1/3$ to keep all previous pivots with the same value.

The fraction operations are put in like $1/3R_1$, $1/3R_3$ and a nice feature of WebCAS is that it uses fractions under the hood and doesn't do a floating-point divide that most languages might do with a $1/3$.

The last pivot that we wanted to do is row 3, column 6. This can be done with

$$\begin{array}{l} -21R_3 + 3R_1 \rightarrow R_1 \\ -14R_3 + 3R_2 \rightarrow R_2 \end{array} \quad \left[\begin{array}{cccccc|c} 27 & 135 & 0 & 0 & 270 & 0 & 1863 \\ 0 & 90 & 27 & 9 & 153 & 0 & 1377 \\ 0 & -6 & 0 & 0 & -12 & -3 & -84 \end{array} \right]$$

Figure 1.4.8 Row operations on the above matrix performed in WebCAS. This is a matrix pivot about row 3, column 6 and the image is the result of the two operations.

and then we'll do the cleanup with

$$\begin{array}{l} -1R_3 \rightarrow R_3 \\ \frac{1}{9}R_1 \rightarrow R_1 \\ \frac{1}{9}R_2 \rightarrow R_2 \end{array} \quad \left[\begin{array}{cccccc|c} 3 & 15 & 0 & 0 & 30 & 0 & 207 \\ 0 & 10 & 3 & 1 & 17 & 0 & 153 \\ 0 & 6 & 0 & 0 & 12 & 3 & 84 \end{array} \right]$$

Figure 1.4.9

and this is the same matrix that we got in [Example 1.3.21](#) and we could then right now the linear system and the solution as we did in that example.

1.4.2 Matrix Pivots in WebCAS

You should clearly notice that using WebCAS to perform the row operations were quite nice in that it eliminates arithmetic errors. However, it still takes a number of steps that can be automated. In this section we will use the `piv` command in WebCAS to further simplify these steps.

Let's return to [Example 1.3.21](#) and if you still have that matrix in the Gaussian Eliminator, then you can click *Restart* and then reenter the matrix.¹ Now at this step if we want to do a matrix pivot about row 1, column 1, then enter `piv(1,1)`, and you should get:

$$\text{piv}(1, 1) \quad \left[\begin{array}{cccccc|c} 3 & 1 & 0 & 0 & 2 & -7 & 11 \\ 0 & 2 & 9 & 3 & -5 & -14 & 67 \\ 0 & 2 & 0 & 0 & 4 & 1 & 28 \end{array} \right]$$

Figure 1.4.10 Pivoting the original matrix about row 1 column 1.

And note that this differs from that of [Figure 1.4.4](#) in that the last two rows differ by a multiple of -1 . One can get from one to the other by row multiplications. The next pivot was about row 2, column 3, so entering $\text{piv}(2, 3)$ results in

$$\text{piv}(2, 3) \quad \left[\begin{array}{cccccc|c} 9 & 3 & 0 & 0 & 6 & -21 & 33 \\ 0 & 2 & 9 & 3 & -5 & -14 & 67 \\ 0 & 6 & 0 & 0 & 12 & 3 & 84 \end{array} \right]$$

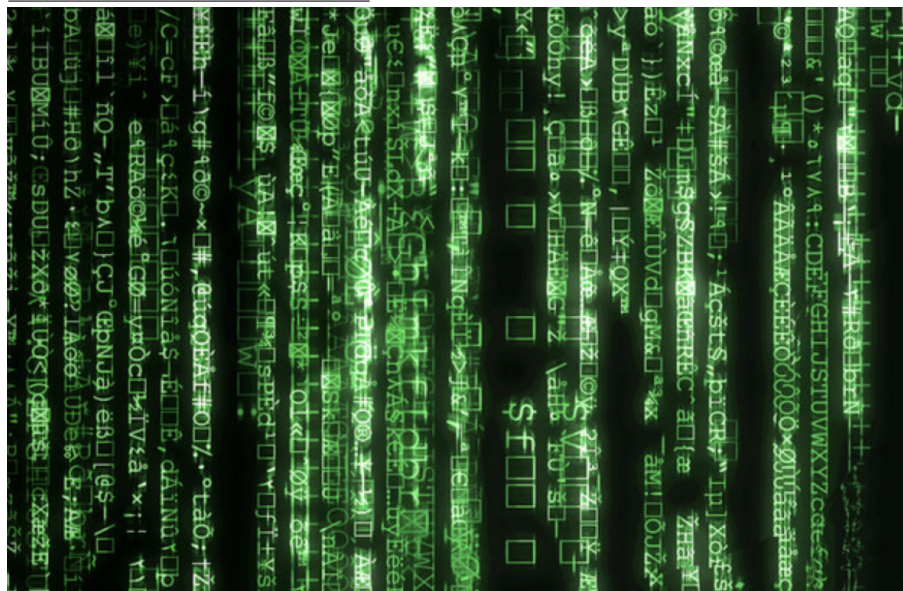
Figure 1.4.11 Pivoting the original matrix about row 2 column 3.

and again this only differs by a sign in the last row in [Figure 1.4.7](#). Lastly, we performed a pivot about row 3, column 6, so $\text{piv}(3, 6)$ results in

$$\text{piv}(3, 6) \quad \left[\begin{array}{cccccc|c} 3 & 15 & 0 & 0 & 30 & 0 & 207 \\ 0 & 10 & 3 & 1 & 17 & 0 & 153 \\ 0 & 6 & 0 & 0 & 12 & 3 & 84 \end{array} \right]$$

Figure 1.4.12 Pivoting the original matrix about row 3 column 6.

and this is now equivalent to the matrix in [Figure 1.4.9](#). Using this fraction-free pivot method, or the `piv` command in WebCAS will keep the matrix in integers and the pivot elements equal (and positive). As we will see, using this and a related form of the pivot will simplify many operations used in later chapters.



1.4.3 Click to Pivot in WebCAS

Another feature of WebCAS is *Click To Pivot*. When selected, this allows the user to click on an element in the matrix and perform the click. Once enabled, you can simply click on any entry in the matrix to perform a pivot about that element. This feature streamlines the process, especially for larger matrices, by reducing the need to manually enter pivot commands. After clicking, the matrix will update automatically to reflect the pivot operation. This will help as the size of the matrix increases.

First, to use this in WebCAS, return to the settings (with the Gear icon) on the top bar, and enable the *Click To Pivot* option then save the changes.

After this is set and a matrix is entered, there will be a Checkbox at the top. Selecting the *Click to Pivot* option puts this in a click mode.

Enter a matrix in the standard way. Then you can click on the number that you wish to pivot upon.